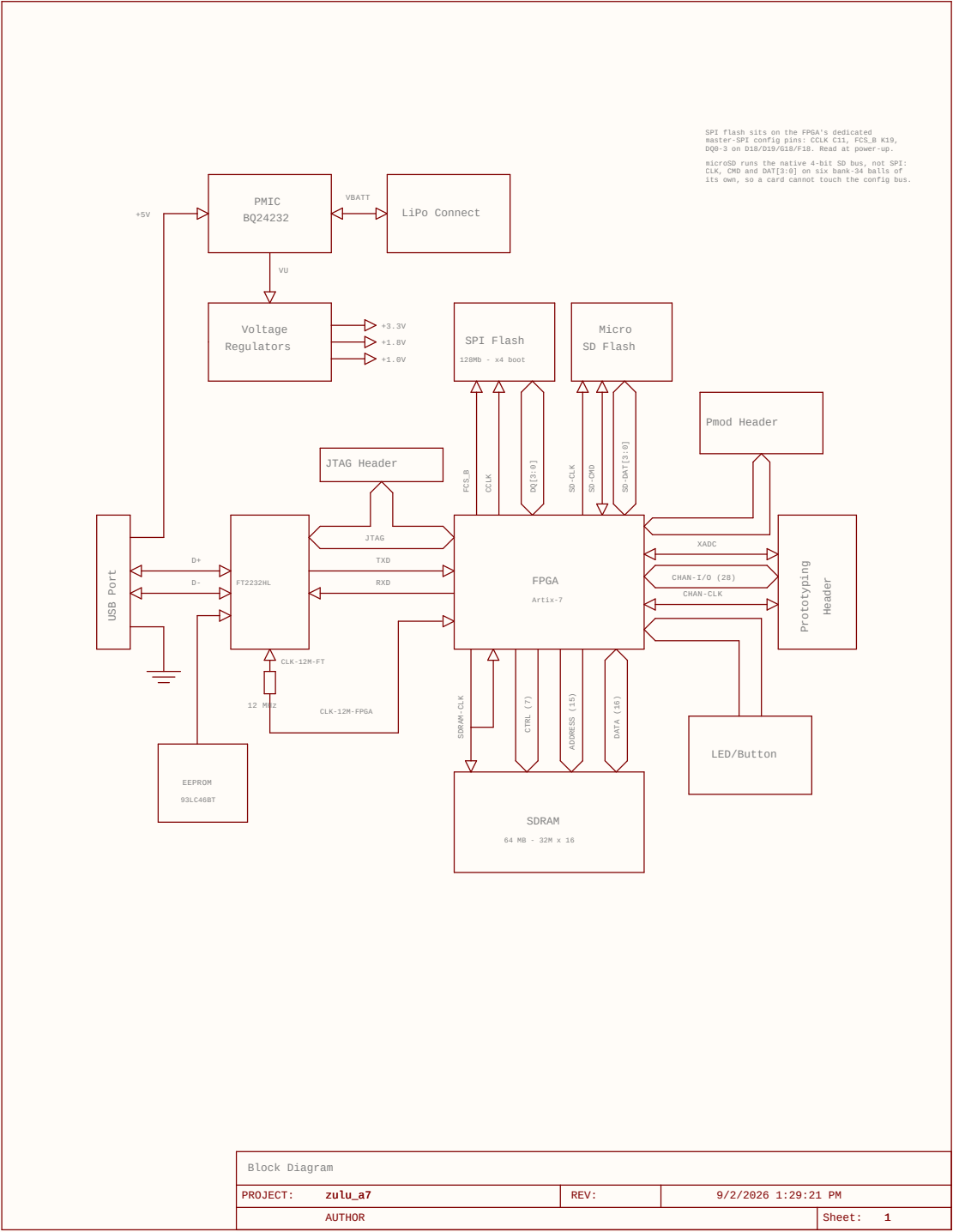
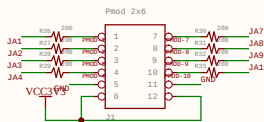
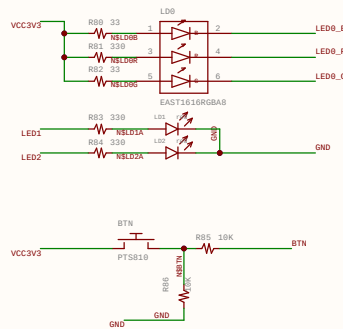
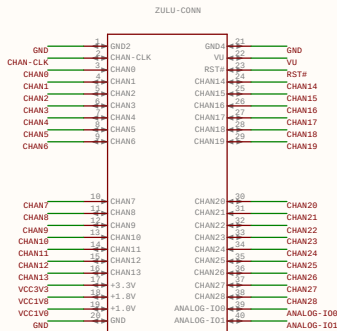


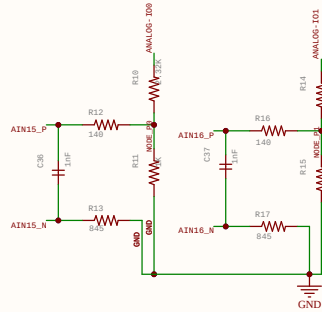




X2: 40 pins on a 2.54 grid, rows 22.86 mm (0.900 in) apart, numbered 1-40 straight through from the right and skipping both landings. The two rows are mirror images. TOP ROW, x 59.69 down to 1.27: GND (pin 1), CHAN-CLK (2), CHAN0-CHAN6 (3-9), then CHAN7, CHAN13 (10-16), +3.3V, +1.8V, +3.3V, GND (17-28), then CHAN7, CHAN13 (29-35), +3.3V, +1.8V, +3.3V, GND (36-40). BOTTOM ROW, CHAN7, CHAN17 (27-29), then CHAN0-CHAN28 (30-38), ANALOG-IO1 (39, 40). Four positions in each row -- x 36.83, 34.29, 31.75, 29.21, between pins 19 and 10 above and between pins 29 and 38 below are RESERVED, not spare. The edge-mounted pins are labeled on the left side of the chip. X1: 40 pins on a 2.54 grid, top row x 7.62 (J28 B2B-PH-5M4 B7, 7.95 mm pin) lands on the bottom row directly opposite it.



Digilent 12-pin Pmod (JA). Pins 1-4 and 7-10 are signals through 200R series resistors; pins 5/11 are GND and 6/12 are VCC3V3 -- so a 2x6 header carries EIGHT signals, not twelve, which is why the names skip JA5/JA6. No decoupling here: the Pmod spec puts bypassing on the module, and VCC3V3 already carries ~170uF. The Cmod A7 has none at its Pmod either. JA1-JA4 (the top row, the half a Pmod uses for SPI) are all in bank 16: A14 A15 B15 A16. JA7 is C15, also bank 16. JA8/JA9/JA10 are W12/W17 -- banks 14/35/34 -- because nothing else was free. All five banks are VCC0-3.3V, so this costs timing coherence, not levels.



U3 = AS4C32M16SB-7TCN: 512Mbit (32M x 16) = 64MB.
Pin-for-pin and dimensionally identical to the 256MB
AS4C16M16SA-6TIN it replaces -- all 54 pins and every
TSP: 1 dimension checked against both datasheets, and
pin 40 is NC on both.
One DIFFERENCE, and IT IS NOT A PIN: the column address
is A0-A9 (1024 columns), not A0-A8 (512). The memory
controller must be set to 10 column bits, or it reaches
only half the array. Row A0-A12, banks BA0/BA1 and
refresh (8192 cycles / 64ms) are unchanged.
-7TCN since 2026-09-08 (-6TIN dry): 143 MHz max at CL13,
tRCD/trp 21, tRC 63 ns, TA 0-70 C, IOD1 max 110 mA.

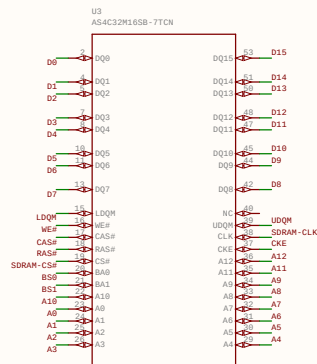
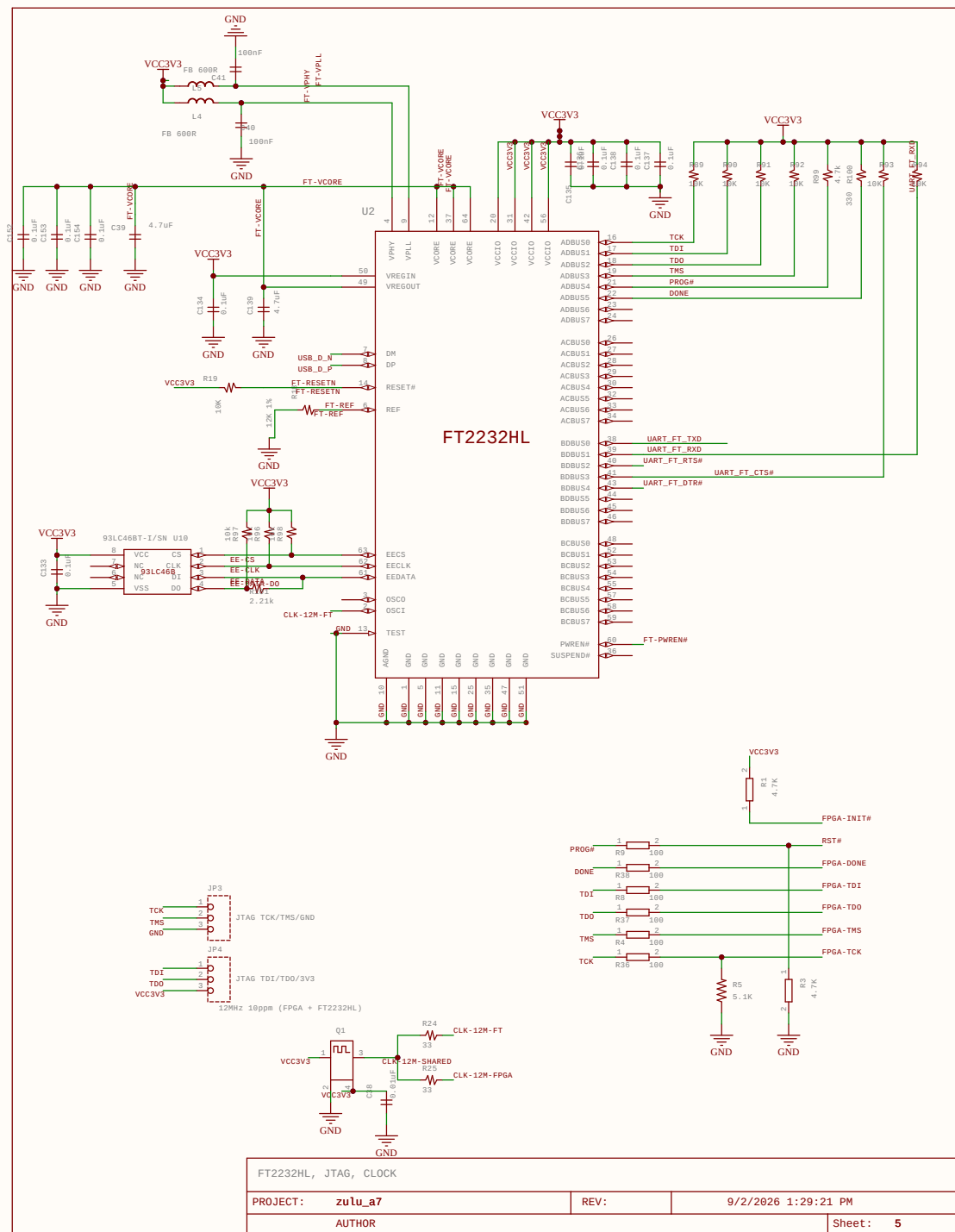
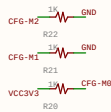


Figure 1 shows the schematic diagram of the proposed 8-bit DAC. The circuit is composed of eight identical stages, each corresponding to a bit of the 8-bit input. The inputs are labeled VSSQ[3], VSSQ[4], VSSQ[5], VSSQ[6], VSSQ[7], VSSQ[8], VSSQ[9], and VSSQ[10]. Each input is connected to a 1-bit DAC core. The outputs of these cores are labeled VDDQ[3], VDDQ[4], VDDQ[5], VDDQ[6], VDDQ[7], VDDQ[8], VDDQ[9], and VDDQ[10]. The circuit uses a 3.3V supply (VCC3V3) and 1.2uF capacitors. The final output is the sum of the individual bit outputs.

Memory			
PROJECT:	zulu_a7	REV:	9/2/2026 1:29:21 PM
AUTHOR			Sheet: 4



PROTOTYPING HEADER



CLK

FLASH



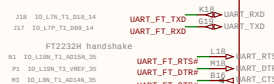
JTAG



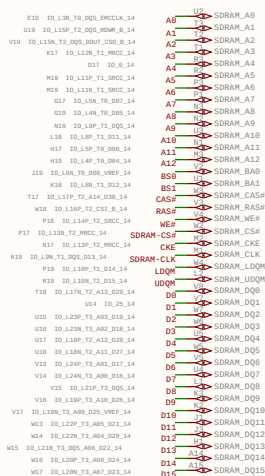
ANALOG



FT2232H



SDRAM



UNCONNECTED BALLS
XADC analog (unused)



Thermal diode (unused)

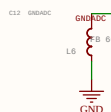


Battery backup (unused)

GTP - float (UG482 TS-4)



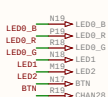
Pmod JA (bank 16 x5)



13 VO



LED / BUTTON



FPGA Connections			
PROJECT:	zulu_a7	REV:	9/2/2026 1:29:21 PM
AUTHOR			Sheet: 6

